High Performance Computing & Data Science Summer Institute 2026

Data Center Tour

Andrea Zonca and Andreas Goetz | August 4, 2026

SDSC
SAN DIEGO SUPERCOMPUTER CENTER

UC San Diego

One briefing, several walk-through times

TUE AUG 4	TUE AUG 4	WED AUG 5	THU AUG 6
11:45 AM-12:15 PM	12:15-12:35 PM	12:00-12:20 PM	12:05-12:25 PM
AUDITORIUM	2 GROUPS	3 GROUPS	3 GROUPS
30-minute presentation	Machine-room walk-throughs	Machine-room walk-throughs	Machine-room walk-throughs

Each participant joins one short walk-through. Five participants plus one guide. Photos are welcome.

The science changed with the machines

1985

~100

researchers applied for time

Astrophysics
Biochemistry
Geology
Oceanography

2016 HANDBOOK

~50%

of computer time reportedly supported life sciences

Engineering and physical sciences remained major users

2026 CONTEXT

208

fields of science in the national ecosystem

900% growth in artificial-intelligence project activity

Examples: genomics, health, materials, wildfire, and climate

The 50 percent figure is historical, not current telemetry.

Forty years of changing machines

1985

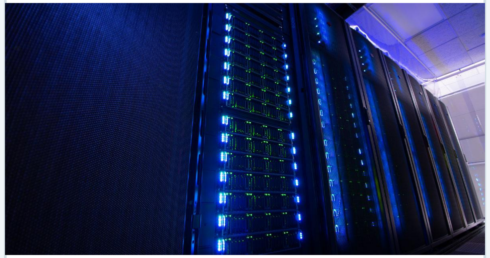
CRAY X-MP/48

FIRST SDSC SYSTEM

Representative photo of the Cray X-MP family

2020

EXPANSE

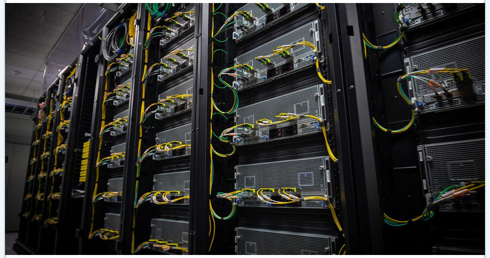


NATIONAL CAPACITY

General-purpose computing plus accelerators

2022

VOYAGER



AI ARCHITECTURE

Purpose-built training and inference hardware

2025

COSMOS

UNIFIED MEMORY

Processors and accelerators share memory

Cray X-MP is a representative public-domain image. Expanse, Voyager, and Cosmos are installed SDSC systems.

U.S. NATIONAL SCIENCE FOUNDATION ACCESS PROGRAM

A national computing ecosystem

ACCESS: Advanced Cyberinfrastructure Coordination Ecosystem: Services & Support

Selected compute providers. ACCESS also integrates storage, cloud, gateways, and support services.

SELECTED COMPUTE-HOSTING PROVIDERS

1 San Diego Supercomputer Center

2 Texas Advanced Computing Center

3 Texas A&M High Performance Research Computing

4 National Center for Supercomputing Applications

5 Pittsburgh Supercomputing Center

6 Purdue University

7 Indiana University

8 U.S. National Science Foundation National Center for Atmospheric Research

9 Stony Brook University

10 University of Kentucky

How a high-performance computing system is organized

USERS + APPS

submit

SCHEDULER

assigns

COMPUTE NODES

connect via

INTERCONNECT

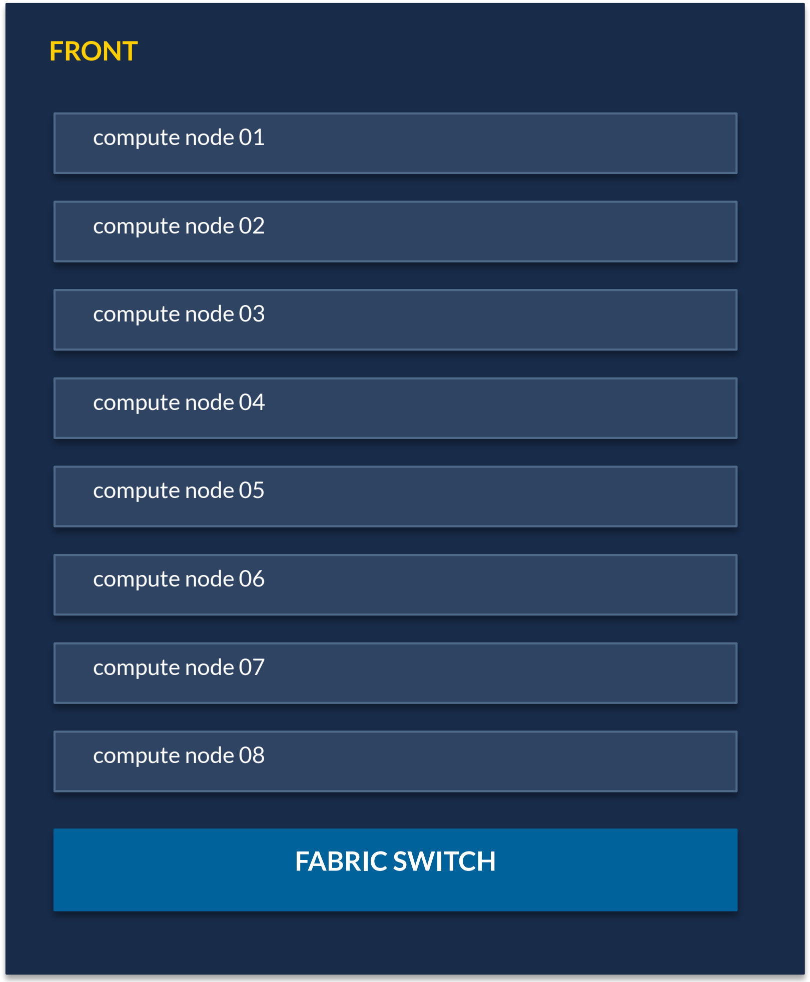
read/write

PARALLEL STORAGE

Compute nodes exchange messages through the interconnect. Parallel storage feeds input and receives checkpoints and results.

FACILITY FOUNDATION: POWER DISTRIBUTION AND COOLING SUPPORT EVERY COMPONENT

How to read a rack



- Repeated server trays: compute nodes
- Dense cable bundles: network and storage fabric
- Vertical power strips: power distribution units (PDUs)
- Perforated doors: air path
- Red and blue hoses: supply and return liquid
- Power distribution units deliver electricity within each rack

Every watt of computing becomes heat

1 kilowatt
ELECTRICAL POWER IN

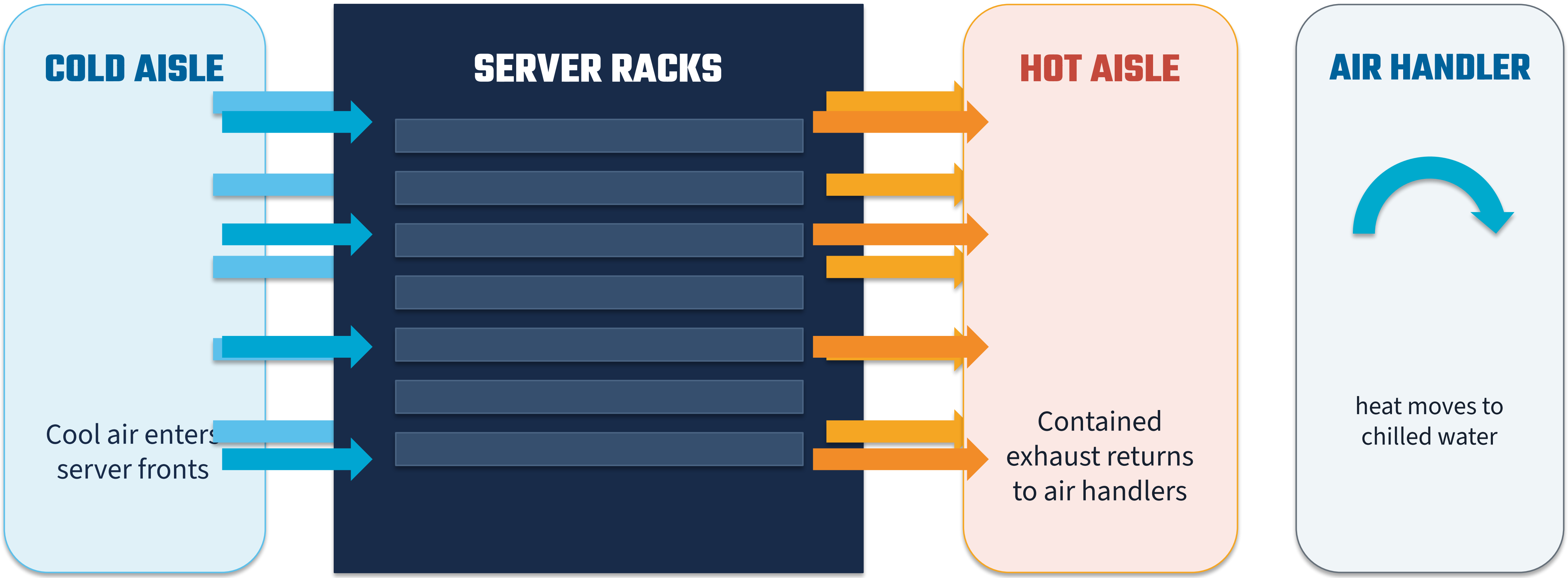
about 1 kilowatt
HEAT TO REMOVE

24/7
COOLING AND MONITORING

- Processors and memory create concentrated heat
- Fans create most of the sound you hear
- Higher rack density pushes designs toward liquid cooling

COOLING PATH 1

Air cooling: keep cold supply and hot exhaust apart



Liquid cooling moves heat closer to the source

PROCESSORS
create heat

›

COLD PLATES
capture heat

›

COOLANT DISTRIBUTION UNIT (CDU)
separates equipment and facility loops

›

FACILITY WATER
carries heat away

2024-2025 FACILITY LIQUID-COOLING UPGRADE

600 kilowatts → 2.4 megawatts

Completed: capacity quadrupled. Liquid distribution now reaches the entire West Data Center.

Cosmos: 100% direct liquid cooled

Power Usage Effectiveness: what supports computing?

PUE

$$\text{PUE} = \frac{\text{TOTAL FACILITY ENERGY}}{\text{COMPUTING EQUIPMENT ENERGY}}$$

1.0 is the theoretical floor

Computing load	processors, network, storage
Cooling	fans, pumps, chillers
Power path	uninterruptible power supply and distribution losses
Facility	lighting and controls

PUE measures facility overhead. It does not tell you whether the scientific code is efficient.

Three machines, three design questions

EXPANSE
SYSTEM IDENTITY

How do we serve many kinds of science?

COMPUTE	General processors + NVIDIA V100 and H100 accelerators
FABRIC	High Data Rate InfiniBand
COOLING	Air plus direct liquid cooling

VOYAGER
SYSTEM IDENTITY

How should hardware for artificial intelligence scale?

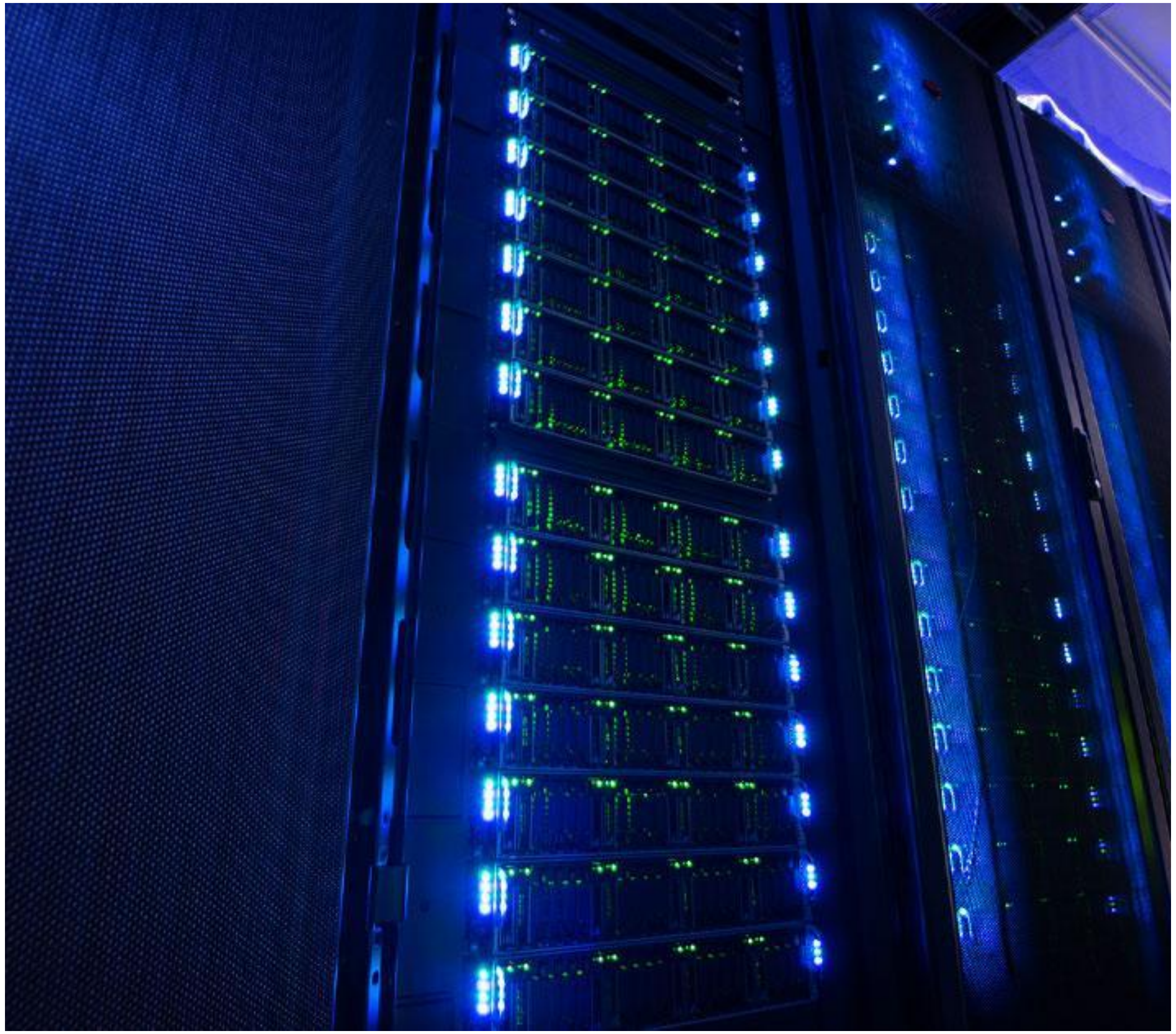
COMPUTE	Gaudi processors
FABRIC	400-gigabit Ethernet
COOLING	Dense rear cabling

COSMOS
SYSTEM IDENTITY

Can shared memory simplify acceleration?

COMPUTE	MI300A unified-memory processors
FABRIC	Slingshot
COOLING	100% direct liquid

Expanse: the national workhorse



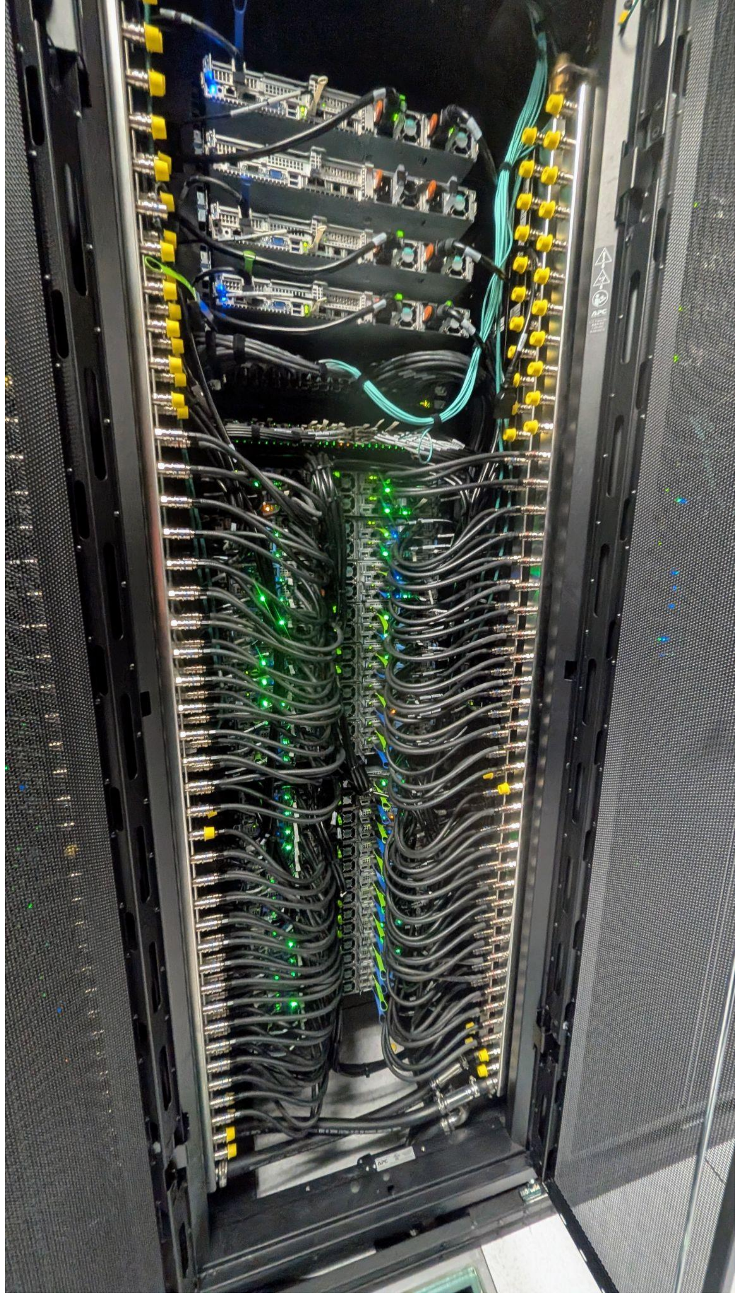
GENERAL-PURPOSE CAPACITY

Many projects, many job sizes

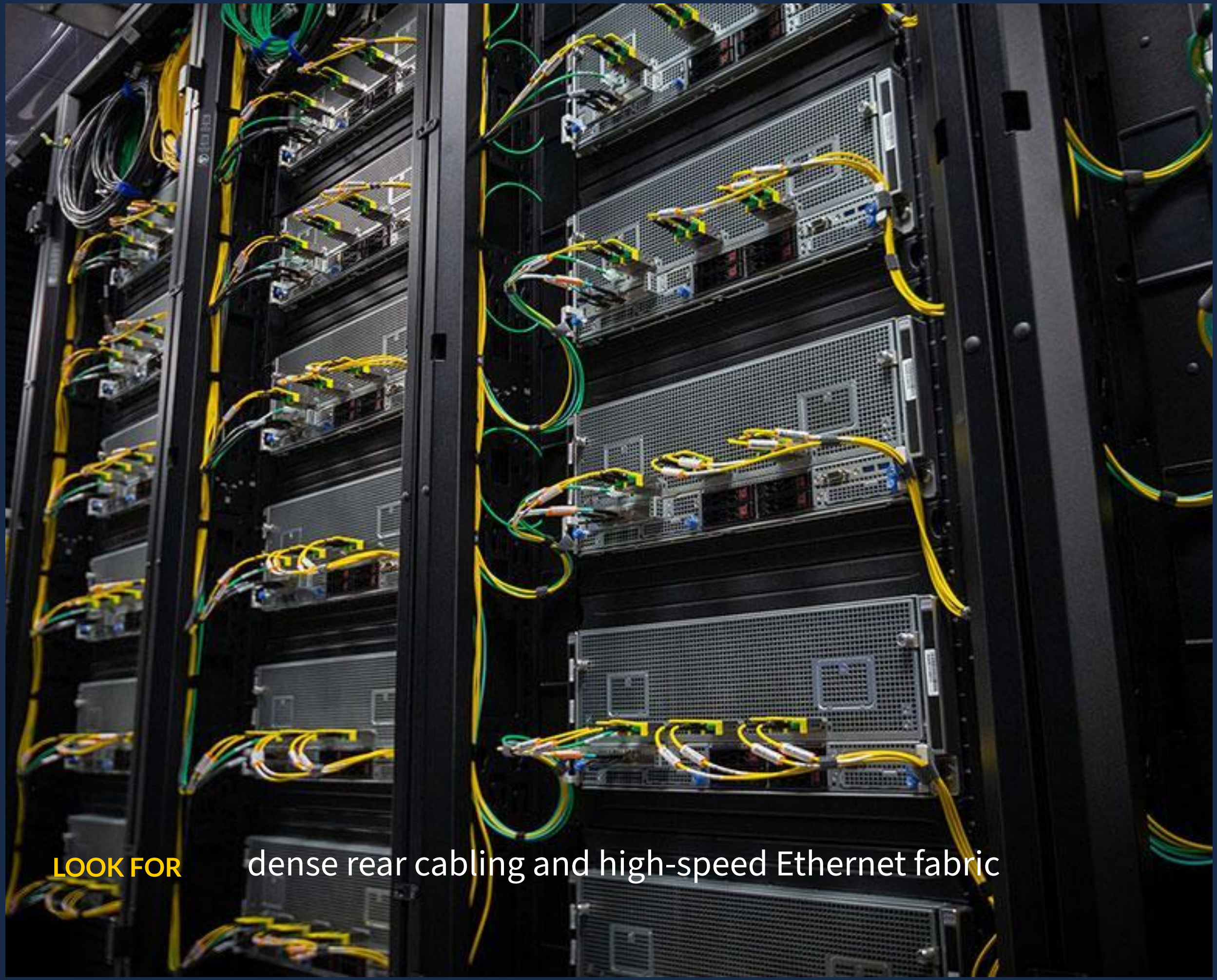
U.S. National Science Foundation awards
2019: $10 million | 2024: $5.3 million expansion

728 STANDARD PROCESSOR NODES	**93,184** PROCESSOR CORES	**5.16** PEAK PETAFLOPS

- 52 NVIDIA V100 accelerator nodes in production
- 34 newer NVIDIA H100 accelerator nodes
- 12 petabytes Lustre plus 7 petabytes Ceph storage
- Direct liquid cooling on original compute nodes



Voyager: artificial-intelligence architecture at scale



TRAINING + INFERENCE

Purpose-built for deep learning

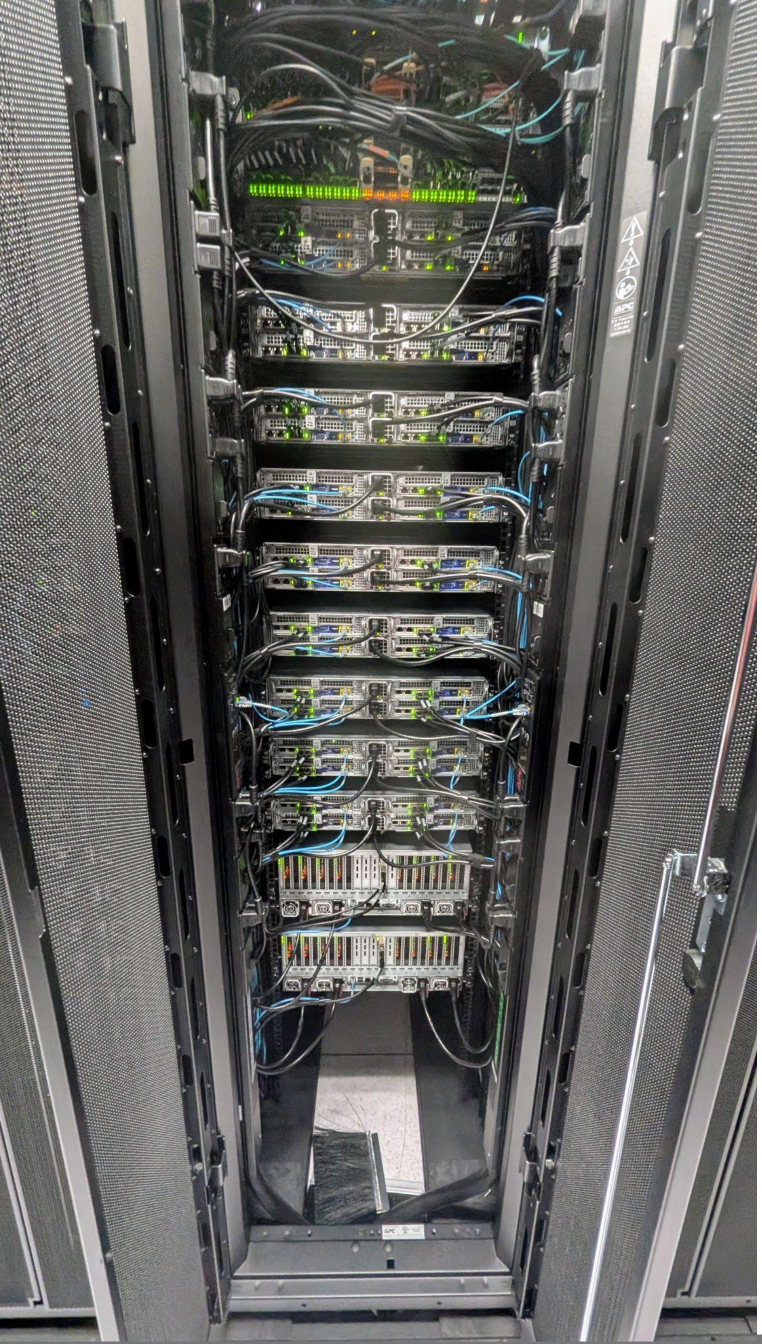
U.S. National Science Foundation
2020: $5 million acquisition award

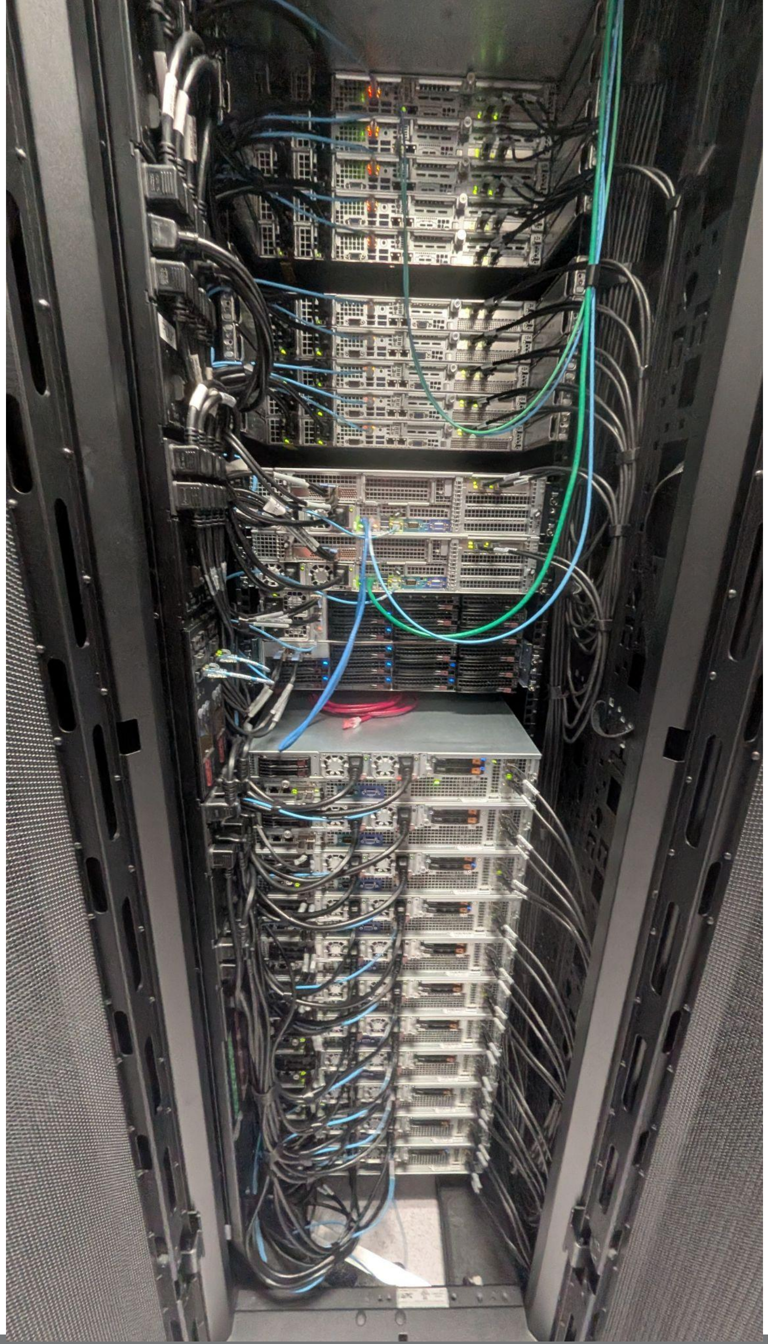
42
GAUDI TRAINING NODES

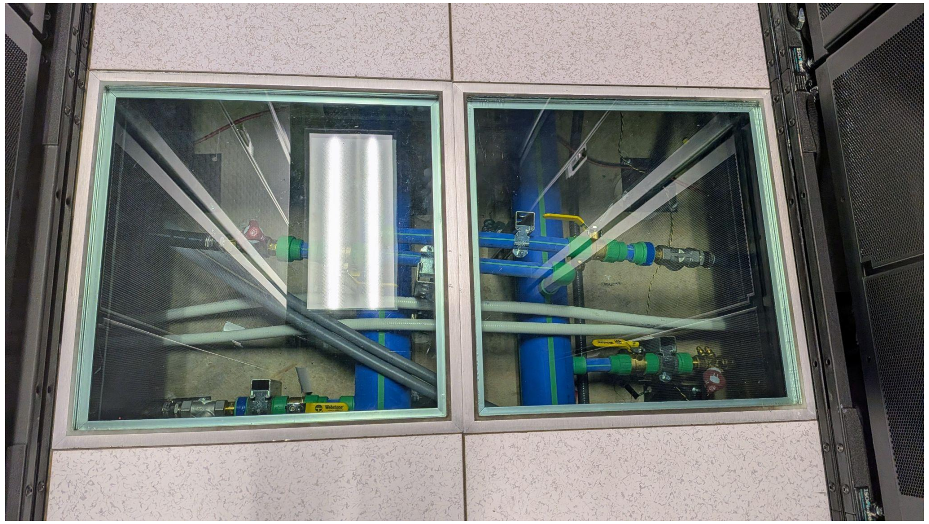
336
GAUDI PROCESSORS

6 x 400
GIGABITS/SECOND LINKS PER NODE

- 8 Gaudi training processors per node
- 16 first-generation inference processors
- Scale-out through an Arista non-blocking Ethernet switch
- PyTorch through the SynapseAI software stack







Cosmos: processor and accelerator share one memory

MI300A ACCELERATED PROCESSING UNIT

24 processor cores + graphics accelerator

128 gigabytes shared high-bandwidth memory

WHY IT MATTERS

Less explicit data copying between processor and accelerator

42
NODES

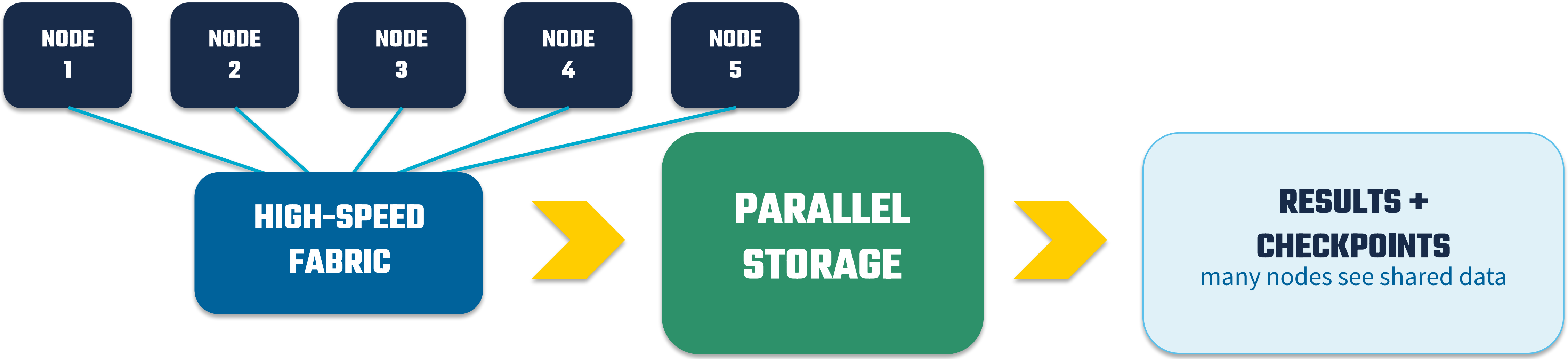
168
MI300A UNITS

200
GIGABITS/SECOND SLINGSHOT

100%
DIRECT LIQUID COOLED

U.S. National Science Foundation | 2024: $5 million initial award
Current phase: testbed

Computation only scales when data can move



Expanse High Data Rate InfiniBand

Main: Lustre + Ceph

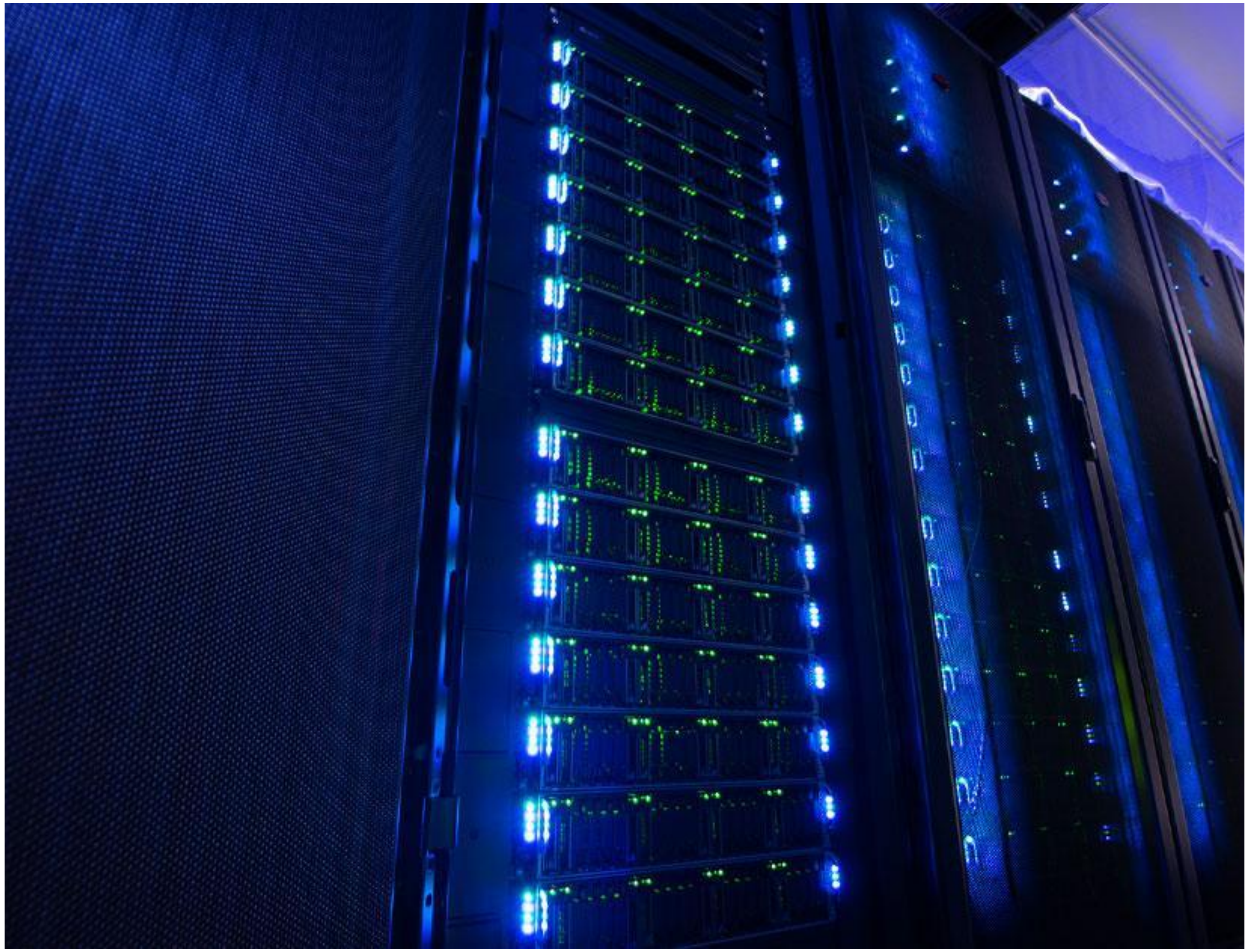
Voyager 400-gigabit Ethernet

Ceph

Cosmos Slingshot interconnect

VAST + Ceph

Your five-minute observation checklist



1
FRONT OR REAR?
Node faces and drive bays usually mark the front. Dense cabling usually marks the rear.

2
AIR OR LIQUID?
Floor tiles or curtains suggest air cooling. Hoses show liquid cooling.

3
COMPUTE OR FABRIC?
Repeated trays compute. Dense cable fields connect.

4
EARTHQUAKE PROTECTION?
Isolation bases permit controlled movement, reducing forces on racks and connections.

Curtains may be touched gently. Do not touch racks, cables, or controls.

Machine-room protocol

5 + 1

Five participants and one center guide per group

STAY CLOSE

Keep the group tight and follow the guide's path

OBSERVE

Minimal talking on the floor; ask questions in the auditorium

NO FOOD

Leave all food and drinks outside the machine room

PHOTOS WELCOME

Take as many photos as you like

CURTAINS OK

Touch curtains gently; hands off equipment, cables, and controls

Full explanations and questions happen in the auditorium.

Ready to recognize what you see?

EXPANSE

capacity workhorse

general processors + NVIDIA accelerators

?

VOYAGER

artificial-intelligence system

Gaudi processors + dense Ethernet fabric

?

COSMOS

unified-memory accelerator

MI300A processors + direct liquid cooling

?

Questions before we enter?

Primary technical sources

1. Center history

   https://www.sdsc.edu/assets/docs/30yr_timeline_final.pdf

2. ACCESS impact metrics

   https://access-ci.org/about/impact/

3. ACCESS resource providers

   https://access-ci.org/resource-providers/

4. Standard system architecture

   https://docs.olcf.ornl.gov/systems/frontier_user_guide.html

5. Expanse user guide

   https://www.sdsc.edu/systems/expanse/user_guide.html

6. Expanse awards

   https://www.sdsc.edu/news/2019/PR20190716_Expanse.html

7. Voyager architecture and award

   https://www.sdsc.edu/news/2020/PR20200701_voyager.html

8. Cosmos user guide and workshop

   https://www.sdsc.edu/systems/cosmos/user_guide.html

9. Cosmos award and liquid cooling

   https://www.sdsc.edu/news/2024/PR20240712_Cosmos_award.html

10. 2024-2025 annual report

    https://www.sdsc.edu/_files/docs/annual-report-2024-web.pdf

Image and template credits

2026 slide master and Welcome design	Cindy Wong and San Diego Supercomputer Center Program and Events
Expanse rack photo	San Diego Supercomputer Center, July 2026
Voyager rack photo	San Diego Supercomputer Center, January 2026
Cosmos cabinet and cooling photos	Mahidhar Tatineni, 2025 Summer Institute Cosmos presentation
Historical image and ACCESS map	Public-domain Cray X-MP photo; current ACCESS catalogs and Bokeh state outlines
Tour narrative	Robert Sinkovits, San Diego Supercomputer Center Tour Handbook, draft 11/30/2016

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